# Problem Set 2

PPHA 34410 Corporate Finance

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## Part 1. Multiple Choice Questions (Select all answers that apply)

**1) According to the Illinois pension article, using a higher assumed discount rate will:**

a. Increase reported pension liabilities
b. Increase required contributions today

**2) Why are overly optimistic pension return assumptions problematic?** (3 marks)

a. They understate liabilities
b. They shift risk to future taxpayers

**3) According to the Nuveen/Schroders article, consolidation in asset management is primarily driven by:**

b. Need for scale
c. Fee compression from passive investing

**4) What types of synergies are expected in the Nuveen–Schroders deal?** (3 marks)

a. Geographic diversification
b. Complementary distribution channels

**5) From the hedge fund weather modeling article, why are firms willing to pay large sums for meteorologists?**

a. Weather directly impacts commodity prices
b. Weather models create arbitrage opportunities
c. Climate volatility increases forecasting value

**6) If SpaceX is added to the S&P 500 shortly after IPO, which of the following are likely consequences?**

c. Index funds will purchase the stock

**7) Based on the "Bonds Are Still Safe" article, key dimensions investors must choose include:**

a. Duration (maturity)
b. Credit quality

**8) According to the Microsoft bond article, why might AAA corporate bonds trade at lower yields than Treasuries?**

b. Extremely strong demand for high-quality corporate bonds

**9) If banning dividends and buybacks leads firms to reinvest more in their core business, why might shareholders still oppose such a policy?**

a. Managers may allocate capital inefficiently
b. Firms may overinvest in negative NPV projects

**10) Which of the following best describes a socially valuable role of dividends and share repurchases?**

b. They recycle capital from mature firms to investors, who can reallocate it to higher-return opportunities elsewhere in the economy

**11) A firm is choosing between two depreciation methods for a new investment:** Method A: Straight-line depreciation over 5 years (capital expenses recognized evenly) Method B: Accelerated depreciation (more expense recognized in earlier years) Assuming the firm is profitable and can fully utilize tax deductions, which of the following best explains why the firm might prefer Method B?

c. It reduces taxes earlier, increasing the present value of tax savings

## Part 2. Short Calculation Question (please show your work to get full credit)

**12) You own the common stock of Grid Batteries. Grid will pay its first annual dividend of $3.00 one year from today. Grid's dividends are predicted to grow by 15% per annum in perpetuity. If this prediction is correct, what is the present value of the dividend stream of this company? Assume the appropriate discount rate is 20% annually.**

Given the assumption that any stock may be thought of as a perpetuity of dividends. Therefore, the equation of growing perpetuity formula is:

$P_0 = \frac{DIV_1}{r-g} = \frac{3}{0.20-0.15} = 60$

**13) I recently purchased a house for $1.4 million. As part of this transaction, I took out a mortgage from a bank where I put down a downpayment equal to 25% of the home value and borrowed the rest of the money from the bank at an interest rate of 6.375%. My expected property taxes on this property are $32,000 per year. What is my leverage ratio?**

Considering that the down payment = equity is 0.25 of the total value of the house, the just dividing:

$\frac{assets}{equity} = \frac{1.4M}{350K} = 4$

Therefore, the leverage ratio is 4x.

**14) I have been raising money from venture capital investment firms for my start-up company, DeepDish Labs. Here is a list of the companies I have raised money from. What is the best estimate for the current post-money valuation of DeepDish Labs?**

The best estimate for the current post-money valuation is the information provided by the latest investment. Therefore, given that Summit Arc bough 5.5% for 3.3M:

$\frac{3.3M}{0.055} = 60M$

The current post-money valuation of the company is 60M.

**15) An auto plant for a new line of magnetic induction autos will cost $1,400 billion to construct. Assume that the plant is available immediately. If successful, the present value of revenues will be $2,400 billion. If unsuccessful, the present value of revenues will be $800 billion. The probability of success is estimated at 35%. We will know immediately after plant completion whether the new auto line will be successful.**

Given the probabilities of success, the expected value is the following:

$E = (0.35 \times 2,400) + (0.65 \times 800) = 840 + 520 = 1,360.$

**a. Would you agree to build the plant?**

$NPV = Revenue - Cost = 1,360 - 1,400 = -40$

No. Given that the cost (1,400 billion) is superior to the expected value of revenue (1,360) I would not agree to build the plant.

**b. Would you agree to build the plant if you knew that you could sell the plant for $900 billion to another automaker immediately after you learn whether the new auto line is successful or unsuccessful?**

Based on the new value for the probability of unsuccessful, the expected value is:

$E = (0.35 \times 2,400) + (0.65 \times 900) = 840 + 585 = 1,425.$

And the NPV:

$NPV = 1,425 - 1,400 = 25.$

Since the NPV now is positive, I would agree to build the plant.